6.1.8 Solve
$$\begin{cases} u_{xx}+u_{yy}+u_{zz}=1 & a<r=\sqrt{x^2+y^2+z^2}<b \quad (i) \\ u=0 & \text{if } r=a \quad (ii) \\ \frac{\partial u}{\partial r}=0 & \text{if } r=b. \quad (iii) \end{cases}$$

We may assume $u=u(r)$ is radially symmetric.
Then by Strauss, page 159, (6),
$u_{xx}+u_{yy}+u_{zz}=1$ is the same as

$$u_{rr}+\frac{2}{r}u_r+\frac{1}{r^2}\left[u_{\theta\theta}+(\cot\theta)u_\theta+\frac{1}{\sin^2\theta}u_{\phi\phi}\right]=1 \quad (1)$$

in spherical coordinates (r,θ,ϕ).
Since by our assumption $u=u(r)$ only depends on r,
(1) is simplified to an ODE in r:

$$u_{rr}+\frac{2}{r}u_r=1. \quad (2)$$

We may solve this ODE, by setting $v\overset{\text{def}}{=}u_r$,

$$v_r+\frac{2}{r}v=1$$

$$\overset{\times r^2}{\Rightarrow} \quad r^2v_r+2r\cdot v=r^2$$

$$\Rightarrow \quad \frac{d}{dr}(r^2\cdot v)=r^2$$

$$\Rightarrow \quad r^2\cdot v=\frac{1}{3}r^3+C_1$$

Since $v(b)=0$ by (iii), we have $C_1=-\frac{1}{3}b^3$.
Thus $v(r)=\frac{1}{3}r-\frac{1}{3}b^3\cdot\frac{1}{r^2}$
Integrating w.r.t. r. then.

$$u(r)=\frac{1}{6}r^2+\frac{1}{3}\cdot b^3\cdot\frac{1}{r}+C_2$$

Since $u(a)=0$ by (ii), we have

$$C_2 = -\frac{1}{6}a^2 - \frac{b^3}{3a}$$

$$\Rightarrow u(r) = \frac{1}{6}r^2 + \frac{b^3}{3r} - \frac{1}{6}a^2 - \frac{b^3}{3a}.$$

i.e.,

$$u(x,y,z) = \frac{1}{6}(x^2+y^2+z^2-a^2) + \frac{b^3}{3}\left(\frac{1}{\sqrt{x^2+y^2+z^2}} - \frac{1}{a}\right).$$

If $a \to 0^+$, $\lim_{a\to 0} u(x,y,z) = -\infty$

(Any reasonable arguments based on that would be good.)

6.1.11 Show that there is no solution of

$$\begin{cases} \Delta u = f & \text{in } D \\ \frac{\partial u}{\partial n} = g & \text{on bdy } D \end{cases} \quad (1)$$

in three dimensions, unless

$$\iiint_D f \, dx\, dy\, dz = \iint_{bdy(D)} g \, dS. \quad (2)$$

Also show the analogue in dim=1 and dim=2.

Proof.

dim=3: It's equivalent to showing if u satisfies (1), then (2) holds.

Since $\Delta f = \operatorname{div}(\operatorname{grad} f)$. we integrate $\Delta u = f$. then

$$\iiint_D \operatorname{div}(\operatorname{grad} u) \, dx\, dy\, dz = \iiint_D f \, dx\, dy\, dz.$$

By divergence thm,

$$\iiint_D \operatorname{div}(\operatorname{grad} u) \, dx\, dy\, dz = \iint_{bdy(D)} \operatorname{grad} u \cdot n \, dS$$

$$(\text{by def'n of } \tfrac{\partial u}{\partial n}) = \iint_{bdy(D)} \frac{\partial u}{\partial n} \, dS$$

$$(\text{by } (1)) = \iint_{bdy(D)} g \, dS$$

dim=1, dim=2 are almost the same, using Stokes' theorem essentially:

If dim=2, the corresponding statement is

If u satisfies $\begin{cases} \Delta u = f \text{ in } D \subset \mathbb{R}^2 \\ \frac{\partial u}{\partial n} = g \text{ on bdy } D \end{cases}$ (1')

~~then~~ $\iint_D f\,dxdy = \oint_{bdy(D)} g\,ds$ (2')

By divergence thm / Green's thm,

$$\iint_D f\,dxdy = \iint_D \Delta u\,dxdy \quad \text{by integrating (1')}$$

$$= \iint_D \operatorname{div}(\operatorname{grad} u)\,dxdy$$

$$= \oint_{bdy(D)} \operatorname{grad} u \cdot \vec{n}\,ds$$

$$= \oint_{bdy(D)} g\,ds$$

If dim $=1$, the corresponding statement is

If u satisfies $\begin{cases} \dfrac{d^2u}{dx^2}=f \text{ in } (a,b)\subset\mathbb{R} & (1'') \\ \dfrac{du}{dx}(a)=A,\ \dfrac{du}{dx}(b)=B, \end{cases}$

where A, B are given constants.

then $\int_a^b f\,dx = B-A$ $(2'')$

By fundemental theorem of calculus,

$$\int_a^b f\,dx = \int_a^b u''\,dx \quad \text{by integrating } (1'')$$

$$= u'(b)-u'(a) \quad \text{by FTC}$$

$$= B-A \quad \text{by } (1'')$$

6.24 Solve $u=u(x,y)$ such that

$$\begin{cases} \Delta u=0 & \text{in } \{0<x<1,\ 0<y<1\} \\ u(x,0)=x & u(x,1)=0 \\ u_x(0,y)=0 & u_x(1,y)=y^2 \end{cases}$$

We'll solve

$$(\mathrm{I})\quad \begin{cases} \Delta u=0 & \text{in } \{0<x<1,\ 0<y<1\} \\ u(x,0)=x \ (1) & u(x,1)=0 \ (2) \\ u_x(0,y)=0 \ (3) & u_x(1,y)=0 \ (4) \end{cases}$$

and $$(\mathrm{II})\quad \begin{cases} \Delta u=0 & \text{in } \{0<x<1,\ 0<y<1\} \\ u(x,0)=0 \ (1') & u(x,1)=0 \ (2') \\ u_x(0,y)=0 \ (3') & u_x(1,y)=y^2. \ (4') \end{cases}$$

and add (I)+(II).

Solve (I): $u(x,y)=X(x)Y(y)$, then there is $\lambda\in\mathbb{R}$ s.t. $X''+\lambda X=0$ for all $0<x<1$

$Y''-\lambda Y=0$ for all $0<y<1$.

By (3) and (4), $X'(0)=X'(1)=0$

(i) $-\lambda = \mu^2 > 0$

$$X = A\cosh(\mu x) + B\sinh(\mu x)$$

$X'(0)=0$. $A\mu\sinh(\mu\cdot 0) + B\mu\cosh(\mu\cdot 0) = 0$

$$\Rightarrow B\mu = 0 \Rightarrow B = 0$$

$X'(1)=0$ $A\mu\sinh(\mu\cdot 1) + B\mu\cosh(\mu\cdot 1) = 0$

$$\Rightarrow A\mu\sinh(\mu) = 0 \underset{\mu\neq 0}{\Rightarrow} A = 0$$

Thus $X(x)\equiv 0$. contradiction.

(ii) $-\lambda = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$X'(0)=0 \Rightarrow A = 0$

$X'(1)=0 \Rightarrow A = 0$

Thus $X\equiv B$ is an eigenfunction w.r.t. $\lambda = 0$,

By $Y'' - \lambda Y = 0$ and $\lambda = 0$. $Y = Cy + D$.

By (2), $B(C+D) = 0 \underset{B\neq 0}{\Rightarrow} C = -D$.

Thus $Y = D(1-y)$

(iii) $-\lambda = -\mu^2 < 0$.

$$X = A\cos(\mu x) + B\sin(\mu x)$$

$X'(0)=0 \Rightarrow -\mu A\sin(\mu\cdot 0) + \mu B\cos(\mu\cdot 0) = 0$

$\Rightarrow \mu B = 0 \Rightarrow B = 0$

$X'(1)=0 \Rightarrow -\mu A\sin(\mu\cdot 1) + \mu B\cos(\mu\cdot 1) = 0$

$\Rightarrow -\mu A\sin\mu = 0$

Since $\mu \neq 0$ and $A \neq 0$, $\sin\mu = 0$, $\Rightarrow \mu = k\pi$, $k \in \mathbb{Z}$.

$$X_n(x) = A_n \cos(n\pi x), \text{ with } \lambda_n := (n\pi)^2.$$

By $Y_n'' - \lambda_n Y_n = 0$ and $\lambda_n = (n\pi)^2$,

$$Y_n(y) = A_n' \cosh(n\pi y) + B_n' \sinh(n\pi y)$$

By (2), $A_n' \cosh(n\pi \cdot 1) + B_n' \cdot \sinh(n\pi \cdot 1) = 0$

$$\Rightarrow Y_n(y) = -\frac{B_n' \cdot \sinh(n\pi)}{\cosh(n\pi)} \cdot \cosh(n\pi y) + B_n' \cdot \sinh(n\pi y)$$

$$= -\frac{B_n'}{\cosh(n\pi)} \left[\sinh(n\pi) \cdot \cosh(n\pi y) - \cosh(n\pi) \sinh(n\pi y)\right]$$

$$= -\frac{B_n'}{\cosh(n\pi)} \cdot \sinh(n\pi - n\pi y) \overset{\text{def}}{=} B_n'' \cdot \sinh(n\pi(1-y)).$$

Thus combining (ii) and (iii) we have a general solution for (I) (without considering (1)) is

$$u(x,y) = BD(1-y) + \sum_{n=1}^{\infty} A_n \cos(n\pi x) \cdot B_n'' \cdot \sinh(n\pi(1-y))$$

$$\overset{\text{def}}{=} E(1-y) + \sum_{n=1}^{\infty} F_n \cdot \cos(n\pi x) \cdot \sinh(n\pi(1-y))$$

We haven't used (1) yet: $u(x,0) = x$ gives

$$x = E \cdot (1-0) + \sum_{n=1}^{\infty} F_n \cdot \sinh(n\pi) \cdot \cos(n\pi x)$$

Note Fourier cosine series of x is $(\ell = 1)$

$$A_0 := \frac{2}{\ell}\int_0^\ell x\,dx = 1$$

$$A_n = \frac{2}{\ell}\int_0^\ell x\cos\left(\frac{n\pi x}{\ell}\right)dx = \begin{cases} -\frac{4}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

By uniqueness of Fourier cosine series, we have

$$E\cdot(1-0) = \frac{A_0}{2} = \frac{1}{2} \Rightarrow E = \frac{1}{2}$$

$$F_n \cdot \sinh(n\pi) = \begin{cases} -\frac{4}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

Hence a solution for (I) is

$$u(x,y) = \frac{1}{2}(1-y)$$

$$+ \sum_{n=1}^{\infty} -\frac{4}{(2n-1)^2\pi^2} \cdot \frac{\sinh((2n-1)\pi)\sinh((2n-1)\pi(1-y))}{\sinh((2n-1)\pi)}$$

Solve (II): $u(x,y) = X(x)Y(y)$, then there is $\lambda \in \mathbb{R}$ s.t. $X'' + \lambda X = 0$ for all $0 < x < 1$

$Y'' - \lambda Y = 0$ for all $0 < y < 1$.

By (1') and (2'), $Y(0) = Y(1) = 0$.

(i) $\lambda = \mu^2 > 0$

$Y = A\cosh(\mu y) + B\sinh(\mu y)$

$Y(0)=0$. $A\cosh(\mu\cdot 0) + B\sinh(\mu\cdot 0) = 0$

$\Rightarrow A = 0$

$Y(1)=0$ $A\cosh(\mu\cdot 1) + B\sinh(\mu\cdot 1) = 0$

$\Rightarrow \sinh(\mu) = 0$

$\Rightarrow \mu = 0$, Contradiction

(ii) $\lambda = 0$, $Y'' = 0 \Rightarrow Y = Ay + B$

$Y(0)=0 \Rightarrow B = 0$

$Y(1)=0 \Rightarrow A+B=0 \Rightarrow A=0$

Thus this case cannot happen.

(iii) $\lambda = -\mu^2 < 0$.

$Y = A\cos(\mu y) + B\sin(\mu y)$

$Y(0)=0 \Rightarrow A\cdot\cos(\mu\cdot 0) + B\cdot\sin(\mu\cdot 0) = 0$

$\Rightarrow A = 0$

$Y(1)=0 \Rightarrow A\cdot\cos(\mu\cdot 1) + B\cdot\sin(\mu\cdot 1) = 0$

$\Rightarrow B\cdot\sin\mu = 0 \Rightarrow \mu = n\pi$ $n\in\mathbb{Z}$.

Thus $Y_n(y) = B_n\cdot\sin(n\pi y)$. $\lambda_n = -(n\pi)^2$

Thus $X_n'' + \lambda_n X_n = 0$ gives

$X_n = A_n'\cdot\cosh(n\pi x) + B_n'\cdot\sinh(n\pi x)$.

By (3') $A_n' \cdot n\pi \cdot \sinh(n\pi \cdot 0) + B_n' \cdot n\pi \cdot \cosh(n\pi \cdot 0) = 0$

$$B_n' \cdot n\pi = 0 \Rightarrow B_n' = 0$$

Thus $X_n(x) = A_n' \cdot \cosh(n\pi x)$

A solution for (II) ignoring (4') is:

$$u(x,y) = \sum_{n=1}^{\infty} A_n' \cdot \cosh(n\pi x) \cdot B_n \sin(n\pi y)$$

$$\overset{\text{def}}{=} \sum_{n=1}^{\infty} B_n'' \cdot \cosh(n\pi x) \cdot \sin(n\pi y).$$

By (4'), $y^2 = \sum_{n=1}^{\infty} B_n'' \sinh(n\pi \cdot 1) \cdot n\pi \cdot \sin(n\pi y)$

Note the Fourier sine series of y^2 over $0<x<1$ is $y^2 = \sum_{n=1}^{\infty} \frac{2}{n^3\pi^3}\left[-2+(-1)^n \cdot (2-n^2\pi^2)\right]\sin(n\pi y).$

By uniqueness,

$$B_n'' \cdot \sinh(n\pi \cdot 1)\, n\pi = \frac{2}{n^3\pi^3}\left[-2+(-1)^n \cdot (2-n^2\pi^2)\right]$$

$$\Rightarrow B_n'' = \frac{2\left[-2+(-1)^n \cdot (2-n^2\pi^2)\right]}{n^4\pi^4 \cdot \sinh(n\pi)}$$

Thus a solution for (II) is

$$u(x,y) = \sum_{n=1}^{\infty} B_n'' \cdot \cosh(n\pi x) \cdot \sin(n\pi y).$$

$$= \sum_{n=1}^{\infty} \frac{2\left[-2+(-1)^n \cdot (2-n^2\pi^2)\right]}{n^4\pi^4 \cdot \sinh(n\pi)} \cdot \cosh(n\pi x) \cdot \sin(n\pi y)$$

Finally, we add the solution for (I) and (II)

$$u(x,y) = \frac{1}{2}(1-y)$$

$$+ \sum_{n=1}^{\infty}\left[-\frac{4}{(2n-1)^2\pi^2} \cdot \frac{\sinh((2n-1)\pi)\sinh((2n-1)\pi(1-y))}{\sinh((2n-1)\pi)} \right.$$

$$\left. + \frac{2[-2+(-1)^n \cdot (2-n^2\pi^2)]}{n^4\pi^4 \cdot \sinh(n\pi)} \cdot \cosh(n\pi x) \cdot \sin(n\pi y) \right]$$

6.2.7. (a) Find the harmonic function in the semi-infinite strip $\{0 \le x \le \pi, 0 \le y < \infty\}$ such that

(*) $u(0,y) = u(\pi,y) = 0$ (1)

$u(x,0) = h(x)$ (2)

$\lim_{y\to\infty} u(x,y) = 0$ (3) (formally, $u(x,\infty) = 0$)

Say $u(x,y) = X(x)Y(y)$,

then $\Delta u = 0$ gives $\frac{X''}{X} = -\frac{Y''}{Y} \overset{\text{def}}{=} -\lambda$

By (1), $X(0) = X(\pi) = 0$.

(i) $-\lambda = \mu^2 > 0$

$X = A\cosh(\mu x) + B\sinh(\mu x)$

$X(0) = 0$. $A\cosh(\mu\cdot 0) + B\cdot\sinh(\mu\cdot 0) = 0$

$\Rightarrow A = 0$

$X(\pi) = 0$ $A\cosh(\mu\cdot\pi) + B\cdot\sinh(\mu\cdot\pi) = 0$

$\Rightarrow \sinh(\mu\cdot\pi) = 0$

$\Rightarrow \mu = 0$. Contradiction

(ii) $-\lambda = 0$, $X_{xx} = 0 \Rightarrow X = Ax + B$

$X(0) = 0 \Rightarrow B = 0$

$X(\pi) = 0 \Rightarrow A\pi + B = 0 \Rightarrow A = 0$

$\Rightarrow X \equiv 0$. Contradiction

(iii) $-\lambda=-\mu^2<0$.

$$X=A\cos(\mu x)+B\sin(\mu x)$$

$$X(0)=0 \Rightarrow A\cos(\mu\cdot 0)+B\cdot\sin(\mu\cdot 0)=0$$
$$\Rightarrow A=0$$
$$X(\pi)=0 \Rightarrow A\cos(\mu\cdot\pi)+B\sin(\mu\cdot\pi)=0$$
$$\Rightarrow \sin(\mu\cdot\pi)=0 \Rightarrow \mu_n\cdot\pi=n\pi \quad n\in\mathbb{Z}$$
$$\Rightarrow \mu_n=n \Rightarrow \lambda_n=n^2$$
$$\Rightarrow X_n=B_n\cdot\sin(nx)$$

Thus $\frac{Y_n''}{Y_n}=\lambda_n=n^2$ gives

$$Y_n=A_n'e^{ny}+B_n'e^{-ny}$$

By $\lim_{y\to\infty}u(x,y)=0$, we have

$$\lim_{y\to\infty}B_n\cdot\sin(nx)(A_n'e^{ny}+B_n'e^{-ny})=0,$$

i.e. $A_n'=0$.

Thus $Y_n=B_n'e^{-ny}$.

A general solution of (*) (without using (2)) is

$$u(x,y)=\sum_{n=1}^{\infty}B_n\sin(nx)\cdot B_n'\cdot e^{-ny}\overset{\text{def}}{=}\sum_{n=1}^{\infty}B_n''\cdot\sin(nx)\cdot e^{-ny}.$$

By (2), $h(x)=\sum_{n=1}^{\infty}B_n''\cdot\sin(nx)$

By uniqueness of Fourier sine series.

$$B_n'' = \frac{2}{\pi}\int_0^{\pi} h(x)\sin(nx)\,dx.$$

Thus a solution of (*)

is $u(x,y) = \sum_{n=1}^{\infty}\left(\frac{2}{\pi}\int_0^{\pi} h(x)\sin(nx)\,dx\right)\cdot \sin(nx)\cdot e^{-ny}$.

(b) What if we omit the condition at ∞?

We cannot get $A_n' = 0$, and it's possible $u(x,y)$ diverges as $y \to +\infty$.

6.3.3 Solve $u=u(x,y)$ such that

$$\begin{cases} u_{xx}+u_{yy}=0 & r<a \\ u=\sin^3\theta & r=a \end{cases}$$

By Strauss P167. (10) (11) (12),

((13) looks good, but is actually hard to compute)

$$u(r,\theta)=\frac{1}{2}A_0+\sum_{n=1}^{\infty} r^n\left(A_n\cos(n\theta)+B_n\sin(n\theta)\right),$$

where $A_n=\frac{1}{\pi a^n}\int_0^{2\pi}\sin^3(\phi)\cos(n\phi)\,d\phi$

$$=\frac{1}{\pi a^n}\int_0^{2\pi}\frac{3\sin(\phi)-\sin(3\phi)}{4}\cdot\cos(n\phi)\,d\phi$$

$$=\frac{1}{4\pi a^n}\int_0^{2\pi}3\sin(\phi)\cos(n\phi)-\sin(3\phi)\cos(n\phi)\,d\phi$$

$$=\frac{1}{4\pi a^n}\int_0^{2\pi}3\cdot\frac{1}{2}\left[\sin(\phi+n\phi)+\sin(\phi-n\phi)\right]$$

$$-\frac{1}{2}\left[\sin(3\phi+n\phi)+\sin(3\phi-n\phi)\right]d\phi$$

$\equiv 0$ for any $n\geq 0$.

$$B_n = \frac{1}{\pi a^n}\int_0^{2\pi} \sin^3(\phi)\sin(n\phi)\,d\phi$$

$$= \frac{1}{\pi a^n}\int_0^{2\pi} \frac{3\sin(\phi)-\sin(3\phi)}{4}\cdot\sin(n\phi)\,d\phi$$

$$= \frac{1}{4\pi a^n}\int_0^{2\pi} 3\sin(\phi)\sin(n\phi)-\sin(3\phi)\sin(n\phi)\,d\phi$$

$$= \frac{1}{4\pi a^n}\int_0^{2\pi} 3\cdot\frac{1}{2}\left[\cos(\phi-n\phi)-\cos(\phi+n\phi)\right]$$

$$-\frac{1}{2}\left[\cos(3\phi-n\phi)-\cos(3\phi+n\phi)\right]d\phi$$

$$= \begin{cases} 0 & \text{if } n\neq 1 \text{ and } n\neq 3 \quad (n\geq 0) \\ \frac{3}{4a} & \text{if } n=1 \\ -\frac{1}{4a^3} & \text{if } n=3 \end{cases}$$

Thus

$$u(r,\theta) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} r^n\left(A_n\cos(n\theta)+B_n\sin(n\theta)\right)$$

$$= r B_1\cdot\sin(\theta) + r^3\cdot B_3\cdot\sin(3\theta)$$

$$= \frac{3r}{4a}\cdot\sin(\theta) - \frac{r^3}{4a^3}\cdot\sin(3\theta)$$